2 Answers to Team Problems

Answer 1. 19

Answer 2. $\frac{5400}{109}$ $\left(\text{or } 49\frac{59}{109}\right)$

Answer 3. $342\sqrt{3}$

Answer 4. $\frac{1}{18}$ $(\text{or } 0.0\overline{5})$

Answer 5. $(2, 3)$

Answer 6. $2\sqrt{73}$

Answer 7. 2,038,181

Answer 8. 132

Answer 9. 20π

Answer 10.

5	1	9	7
5	1	0	7
4	1	0	2
4	8	0	2

3 Solutions to Team Problems

Problem 1. Compute the number of non-empty subsets S of $\{1, 2, 3, \ldots, 20\}$ for which $|S| \cdot \max\{S\} = 18$. (Note: $|S|$ is the number of elements of the set S.)

Solution 1. Both $|S|$ and $\max\{S\}$ are integers between 1 and 20, so factor 18 and consider the possible ways to have $|S|$ and $\max\{S\}$ equal each factor. Then count the number of possible sets S that satisfy that condition.

$\boldsymbol{|S|}$	$\boldsymbol{\max\{S\}}$	**Number of possible sets S**
1	18	1: this can only be the set $S = \{18\}$.
2	9	8: this can be any set $S = \{a, 9\}$, where $1 \leq a \leq 8$.
3	6	$\binom{5}{2} = 10$: the greatest element must be 6, and there are $\binom{5}{2}$ ways of selecting the two other elements of S.
6	3	0: this is impossible because S cannot have 6 elements, the greatest of which is 3.
9	2	0: this is impossible because S cannot have 9 elements, the greatest of which is 2.
18	1	0: this is impossible because S cannot have 18 elements, the greatest of which is 1.

Thus the answer is $1 + 8 + 10 = \mathbf{19}$.

Problem 2. A class of 218 students takes a test. Each student's score is an integer from 0 to 100, inclusive. Compute the greatest possible difference between the mean and the median scores.

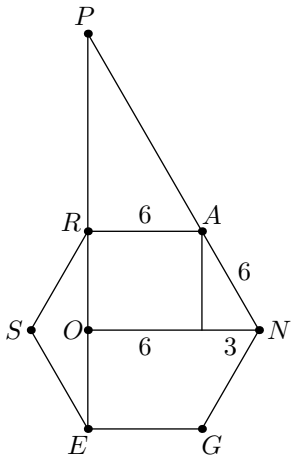
Solution 2. Intuitively, the maximum difference can be attained by forcing the median to be 0 by selecting just enough 0 scores, and then maximizing the mean by maximizing the remaining scores. Such a distribution would have 110 scores of 0 and 108 scores of 100, giving a median score of 0 and a mean score of $\frac{108 \cdot 100}{218} = \frac{5400}{109}$. This gives a difference of $\frac{5400}{109}$.

To prove that this is maximal, assume that the median score is m. Then, by definition, at least 109 scores must be at most m. Further, the remaining 109 scores are at most 100, so the mean μ satisfies $\mu \leq \frac{1}{218}(109m + 10900) = \frac{m+100}{2}$. Therefore $\mu - m \leq \frac{100-m}{2}$.

If $m = 0$, then the maximum mean is found as above. If $m \geq 1$, then $\mu - m \leq \frac{100-1}{2} = 49.5 < \frac{5400}{109}$, and the difference is greater than it is in the above case with $m = 0$. There only remains the possibility that $m = \frac{1}{2}$. For this to be the case, 109 scores must be 0 and one score must be 1. To maximize the mean, the remaining 108 scores must be 100. However, compared to the case of having 110 scores of 0 and 108 of 100, this increases the median by $\frac{1}{2}$ but increases the mean by only 1218, so the difference will certainly be smaller. Thus the answer is $\mathbf{\frac{5400}{109}}$.

Problem 3. Regular hexagon $RANGES$ has side length 6. Pentagon $RANGE$ is revolved 360° about the line containing $\overline{RE}$ to obtain a solid. The volume of the solid is $k \cdot \pi$. Compute k.

Solution 3. First note that because $\overline{RA}$ and $\overline{EG}$ are parallel, the solid obtained will consist of two conic frustums, each with a base along the plane through $\overline{NS}$ and perpendicular to $\overline{ER}$. It is easier to compute the volume of one of these conic frustums at a time. Let O be the midpoint of $\overline{ER}$, and let $\overrightarrow{NA}$ and $\overrightarrow{ER}$ intersect at P. Then the conic frustum is a cone with radius ON and height OP, cut off by the plane containing $\overline{AR}$ and perpendicular to $\overline{ER}$.



Note that $ON = NS - OS = 12 - 3 = 9$, and $OP = OR + RP = 3\sqrt{3} + 6\sqrt{3} = 9\sqrt{3}$. Then the volume of one of the conic frustums is

$$\frac{1}{3}\pi\left(9^2\right)\left(9\sqrt{3}\right) - \frac{1}{3}\pi\left(6^2\right)\left(6\sqrt{3}\right) = 171\sqrt{3}\,\pi.$$

The volume of the entire solid is therefore $342\sqrt{3}\cdot\pi$, so $k = \mathbf{342\sqrt{3}}$.

Problem 4. A fair 12-sided die has faces numbered 1 through 12. The die is rolled twice, and the results of the two rolls are x and y, respectively. Given that $\tan(2\theta) = \frac{x}{y}$ for some θ with $0 < \theta < \frac{\pi}{2}$, compute the probability that $\tan\theta$ is rational.

Solution 4. Suppose $\tan 2\theta = \frac{x}{y}$. Then the identity for the tangent of a sum gives $\frac{2\tan\theta}{1-\tan^2\theta} = \frac{x}{y}$, which is equivalent to $x - x\tan^2\theta = 2y\tan\theta$. Rearranging gives a quadratic equation in $\tan\theta$: $x\tan^2\theta + 2y\tan\theta - x = 0$. Thus $\tan\theta$ is rational if and only if the discriminant is a perfect square. The discriminant is $(2y)^2 - 4\cdot x\cdot(-x) = 4(x^2+y^2)$, which is a perfect square if and only if x^2+y^2 is a perfect square. The only Pythagorean triples (a,b,c) with $a,b \le 12$ are $(3,4,5),(6,8,10),(9,12,15)$, and $(5,12,13)$. Therefore the only (x,y) that give a rational value for $\tan\theta$ are $(3,4),(4,3),(6,8),(8,6),(9,12),(12,9),(5,12),(12,5)$. The desired probability is therefore $\frac{8}{12\cdot 12} = \mathbf{\frac{1}{18}}$.

Problem 5. The absolute values of the five complex roots of $z^5 - 5z^2 + 53 = 0$ all lie between the positive integers a and b, where $a < b$ and $b - a$ is minimal. Compute the ordered pair (a, b).

Solution 5. First note that the question asks for the tightest possible integer bounds on the magnitude of a zero of the complex polynomial $z^5 - 5z^2 + 53$, essentially describing the smallest complex annulus with integer radii that contains the roots.

Now recall two important properties of the absolute value function: $|ab| = |a||b|$ and $|a+b| \le |a| + |b|$ for all complex numbers a and b. The latter also implies that $|a| - |b| \le |a-b|$. In order to have $z^5 - 5z^2 + 53 = 0$, it must be the case that $|z^5 - 5z^2| = 53$. This is enough to restrict the possible magnitude of z.

Suppose that $|z| > 3$. Then $|z^5 - 5z^2| = |z|^2|z^3 - 5| > 9|z^3 - 5| \ge 9(|z|^3 - 5) > 9(27 - 5) = 198 > 53$. So $|z| \le 3$.

Suppose that $|z| < 2$. Then $|z^5 - 5z^2| = |z|^2|z^3 - 5| < 4|z^3 - 5| \le 4(|z|^3 + |5|) < 4(8 + 5) = 51 < 53$. So $|z| \ge 2$.

Therefore $2 \le |z| \le 3$ for all roots z of $z^5 - 5z^2 + 53$. Because $a < b$, and a and b are integers, $b - a$ is at least 1, so this is the best possible bound. The answer is $\mathbf{(2, 3)}$.

Problem 6. Let S be the set of points (x, y) whose coordinates satisfy the system of equations:

$$\lfloor x \rfloor \cdot \lceil y \rceil = 20$$
$$\lceil x \rceil \cdot \lfloor y \rfloor = 18.$$

Compute the least upper bound of the set of distances between points in S.

Solution 6. First note that x and y cannot both be integers because otherwise, the given system would be inconsistent. Now suppose that exactly one of x or y is an integer. If x is an integer and $\lfloor y \rfloor = b$, then $\lceil y \rceil = b + 1$, and multiplying the two given equations and simplifying results in the equation $x^2 \cdot b(b+1) = 360$ (†). Because $1 \leq x^2 < 360$, the only possible values of x^2 are the perfect square factors of 360, namely 1, 4, 9, and 36. Substituting in $x^2 = 1, 9$, and 36 into (†) results in non-integral solutions for b. But substituting $x^2 = 4$ into (†) results in $b^2 + b - 90 = (b + 10)(b - 9) = 0$, and $b = -10$ or $b = 9$. If $x = 2$, then $\lfloor y \rfloor = 9$ and $\lceil y \rceil = 10$, and this yields the solutions $(2, y)$, where $9 < y < 10$. On the other hand, if $x = -2$, then $\lfloor y \rfloor = -9$ and $\lceil y \rceil = -10$, which is impossible. By a similar argument, if x is not an integer and y is an integer, this results in the solutions $(x, -2)$, where $-10 < x < -9$. Thus if $\mathcal{I}_1$ is the open interval $(9, 10)$ and $\mathcal{I}_2$ is the open interval $(-10, -9)$, then the solutions (x, y) to the given system where exactly one of x and y is an integer are the ordered pairs belonging to the union of the two sets $\{2\} \times \mathcal{I}_1$* and $\mathcal{I}_2 \times \{-2\}$. Graphically, these represent segments of unit length (one vertical, one horizontal) that do not include the endpoints. For later reference, let $\mathcal{I}_x = \mathcal{I}_2 \times \{-2\}$ and let $\mathcal{I}_y = \{2\} \times \mathcal{I}_1$.

Now suppose that neither x nor y is an integer and that $\lfloor x \rfloor = a$ and $\lfloor y \rfloor = b$. Then $\lceil x \rceil = a + 1, \lceil y \rceil = b + 1$, and multiplying the two given equations and simplifying results in the equation $a(a+1)b(b+1) = 360$ (‡). By noting the factorization $360 = 3 \cdot 4 \cdot 5 \cdot 6$, the following ordered pairs (a, b) satisfy (‡):

$$(3, 5); \quad (-4, 5); \quad (-4, -6); \quad (3, -6); \quad (5, 3); \quad (5, -4); \quad (-6, -4); \quad (-6, 3).$$

However, only $(a, b) = (5, 3)$ and $(-4, -6)$ satisfy the given system:

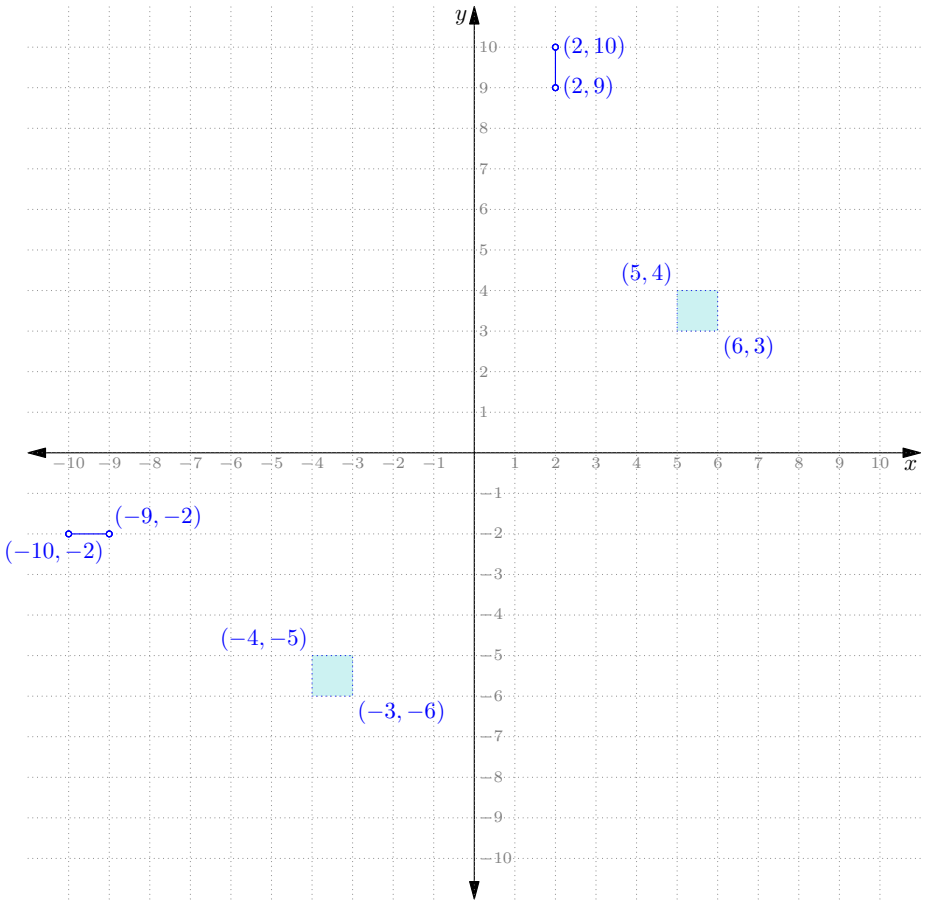
$$a(b + 1) = 20$$
$$(a + 1)b = 18.$$

Let $\mathcal{I}_3$ and $\mathcal{I}_4$ denote the open intervals $(5, 6)$ and $(3, 4)$, respectively and let $\mathcal{I}_5$ and $\mathcal{I}_6$ denote the open intervals $(-4, -3)$ and $(-6, -5)$, respectively. Then the solutions (x, y) to the given system in which neither x nor y is an integer are given by the union of the two regions $\mathcal{R}_1$ and $\mathcal{R}_2$, defined by

$$\mathcal{R}_1 = \mathcal{I}_3 \times \mathcal{I}_4 \quad \text{and} \quad \mathcal{R}_2 = \mathcal{I}_5 \times \mathcal{I}_6.$$

Each of $\mathcal{R}_1$ and $\mathcal{R}_2$ is the interior of a unit square ($\mathcal{R}_1$ lies in the first quadrant and $\mathcal{R}_2$ lies in the third quadrant). Also note that the solutions are symmetric about the line $y = -x$. A plot of the points of S is shown.

*Here, $\times$ denotes the Cartesian product. I.e., if P and Q are two sets, then $P \times Q = \{(p, q) \mid p \in P \text{ and } q \in Q\}$.



The answer to the problem is $\max\{\ell_1, \ell_2, \ell_3\}$, where:

- ℓ_1 is the least upper bound of the distances between a point in $\mathcal{I}_x$ and a point in $\mathcal{I}_y$;
- ℓ_2 is the least upper bound of the distances between a point in $\mathcal{R}_1$ and a point in $\mathcal{R}_2$;
- ℓ_3 is the least upper bound of the distances between a point in $\mathcal{I}_x \cup \mathcal{I}_y$ and a point in $\mathcal{R}_1 \cup \mathcal{R}_2$.

Compute ℓ_1 by taking the distance between the extremal boundary points of $\mathcal{I}_x$ and $\mathcal{I}_y$, namely $(-10, -2)$ and $(2, 10)$, respectively. This distance is $\sqrt{2 \cdot 12^2} = \sqrt{288}$.

Compute ℓ_2 by taking the distance between the extremal boundary points of $\mathcal{R}_1$ and $\mathcal{R}_2$, namely $(6, 4)$ and $(-4, -6)$, respectively. This distance is $\sqrt{2 \cdot 10^2} = \sqrt{200}$.

Compute ℓ_3 by taking the distance between the extremal boundary points of $\mathcal{I}_x$ and $\mathcal{R}_1$ (or of $\mathcal{I}_y$ and $\mathcal{R}_2$, owing to the symmetry), namely $(-10, -2)$ and $(6, 4)$, respectively. This distance is $\sqrt{16^2 + 6^2} = \sqrt{292}$.

Thus the answer is $\max\{\sqrt{288}, \sqrt{200}, \sqrt{292}\} = \sqrt{292} = \mathbf{2\sqrt{73}}$.

Problem 7. Compute the least integer $d > 0$ for which there exist distinct lattice points A, B, and C in the coordinate plane, each exactly $\sqrt{d}$ units from the origin, satisfying $\csc(\angle ABC) > 2018$.

Solution 7. The points A, B, and C lie on a circle with diameter $2\sqrt{d}$. Consequently, by the Extended Law of Sines,

$$2\sqrt{d} = \frac{AC}{\sin(\angle ABC)} \quad \rightarrow \quad \csc(\angle ABC) = \frac{2\sqrt{d}}{AC}\,.$$

The solution now proceeds in three cases.

- It is impossible that $AC = 1$. Indeed, if $A = (x, y)$, then without loss of generality, assume $C = (x+1, y)$. Then $d = x^2 + y^2 = (x+1)^2 + y^2$, which implies $2x + 1 = 0$, contradiction.
- Suppose $AC = \sqrt{2}$. If $A = (x, y)$, then without loss of generality, assume $C = (x + 1, y + 1)$, by reflecting or rotating as necessary. Then $d = x^2 + y^2 = (x + 1)^2 + (y + 1)^2$, hence $2x + 1 + 2y + 1 = 0$. It follows that $y = -(x + 1)$ and d must be of the form $d = x^2 + (x + 1)^2$. In that case,

$$\csc(\angle ABC) = \frac{2\sqrt{d}}{\sqrt{2}} = \sqrt{2(x^2 + (x+1)^2)}$$

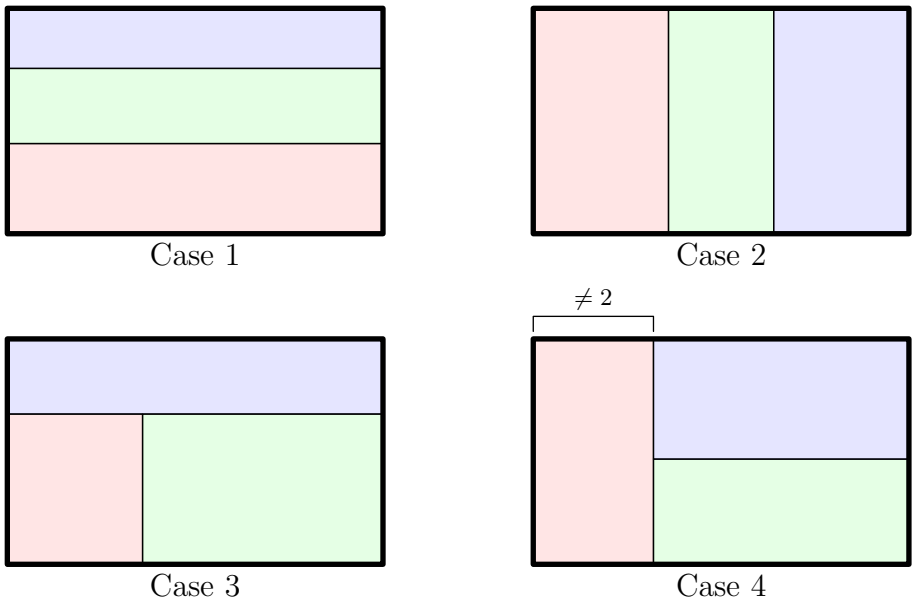
  and the least x for which this exceeds 2018 is $x = 1009$, meaning $d = 1009^2 + 1010^2$.
- Finally assume $AC \geq \sqrt{3}$. Then $2018 < \frac{2\sqrt{d}}{AC} \leq \sqrt{\frac{4}{3}d}$, which would mean $d > 2018^2 \cdot \frac{3}{4} > 1009^2 + 1010^2$. Thus all values of d achieved in this case are greater than the value of d in the second case.

In conclusion, the least possible value of d is $1009^2 + 1010^2 = \mathbf{2038181}$. This value of d is indeed possible, as the points $A = (1009, 1010)$, $B = (-1009, -1010)$, and $C = (1010, 1009)$ satisfy the conditions of the problem.

Problem 8. Compute the number of unordered collections of three integer-area rectangles such that the three rectangles can be assembled without overlap to form one 3×5 rectangle. (For example, one such collection contains one 3×3 and two 1×3 rectangles, and another such collection contains one 3×3 and two 2×1.5 rectangles. The latter collection is equivalent to the collection of two 1.5×2 rectangles and one 3×3 rectangle.)

Solution 8. In order to enumerate the possible rectangles it is necessary to consider several cases. One possible approach is presented below.

Consider the possible dissections of the given 3×5 rectangle. There are four shapes which can be achieved, as shown below.



Begin by counting the number of unordered collections that arise from each case.

- In Case 1, a collection corresponds to a multiset $\{a, b, c\}$ satisfying $a + b + c = 15$ (as the three rectangles can be reordered freely). Assume, without loss of generality, that $a \leq b \leq c$. To eliminate the inequality

constraints, let $x = a - 1$, $y = b - a$, $z = c - b$ denote nonnegative integers. Accordingly, $a = x + 1$, $b = x + y + 1$, $z = x + y + z + 1$, and the given equation now reads $x + 2y + 3z = 12$ or, equivalently, $2y + 3z \leq 12$. The number of solutions in nonnegative integers is then

$$\sum_{z=0}^{4} \left(1 + \left\lfloor \frac{12 - 3z}{2} \right\rfloor\right) = 7 + 5 + 4 + 2 + 1 = 19.$$

- Case 2 has 19 solutions by repeating the argument from Case 1 verbatim.
- In Case 3, denote by $15 - n$ the area of the top rectangle. The number of possible dissections of the lower rectangle is then exactly $\lfloor n/2 \rfloor$. Moreover, the top rectangle (because it has a side length of 5) is never congruent to either of the lower rectangles. Thus the number of solutions in this case is

$$\sum_{n=1}^{14} \left\lfloor \frac{n}{2} \right\rfloor = 49.$$

- In Case 4, assume the leftmost region does not have width 2. This ensures that Case 2 and Case 4 will be disjoint from each other, because the two rectangles on the right will not form a 3×3 square that could be rotated to obtain a dissection already counted in Case 2.

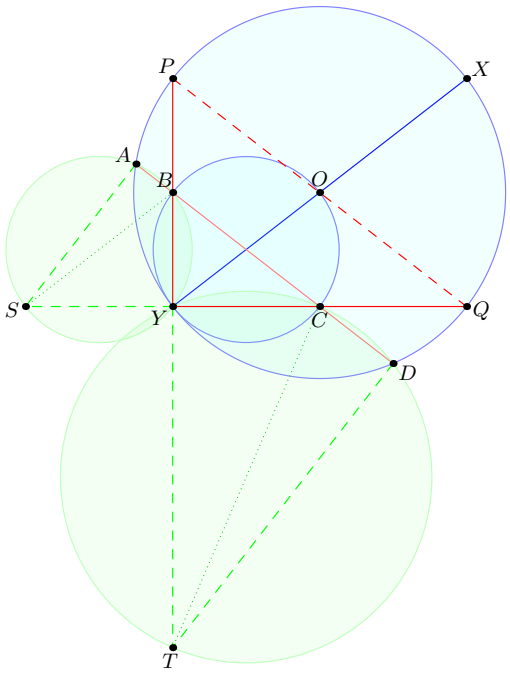
  In order to count the number of such configurations, denote by $15 - n$ the area of the left rectangle, where $15 - n \neq 3 \cdot 2$ or, equivalently, $n \neq 9$. Then repeating the logic of Case 3 gives a count of

$$\sum_{\substack{n=1 \\ n \neq 9}}^{14} \left\lfloor \frac{n}{2} \right\rfloor = 49 - \left\lfloor \frac{9}{2} \right\rfloor = 45.$$

It has already been checked that Cases 3 and 4 are disjoint, and the other pairs of cases are seen to be mutually exclusive by comparing the number of sides of length 5 in the dissection. Consequently, the final answer is $19 + 19 + 49 + 45 = \mathbf{132}$.

Problem 9. Let Γ be a circle with diameter $\overline{XY}$ and center O, and let γ be a circle with diameter $\overline{OY}$. Circle ω_1 passes through Y and intersects Γ and γ again at A and B, respectively. Circle ω_2 also passes through Y and intersects Γ and γ again at D and C, respectively. Given that $AB = 1$, $BC = 4$, $CD = 2$, and $AD = 7$, compute the sum of the areas of ω_1 and ω_2.

Solution 9. Note that A, B, C, and D are collinear, in that order, because $AD = AB + BC + CD$. Extend $\overline{YB}$ and $\overline{YC}$ to meet Γ again at P and Q, respectively. Then $\overline{BC}$ is the midline of $\triangle YPQ$, by homothety. By power of a point, $AB \cdot BD = 1 \cdot (4+2) = PB \cdot BY = BY^2$, so $BY = \sqrt{6}$. Similarly, $CY = \sqrt{2 \cdot (4+1)} = \sqrt{10}$. In particular, $\triangle BYC$ is right.



Now let S and T be the antipodes of B and C on ω_1 and ω_2, respectively. It remains to evaluate BS and CT. First note that $\mathrm{m}\angle BYS = 90^\circ$ and $\mathrm{m}\angle CAS = \mathrm{m}\angle BAS = 90^\circ$. Because $\mathrm{m}\angle BYC = 90^\circ$, it follows that points C, Y, and S are collinear and that $\triangle CYB \sim \triangle CAS$. Therefore

$$AS = \frac{YB}{YC} \cdot AC = \sqrt{\frac{6}{10}} \cdot 5 = \sqrt{15},$$

hence $BS = \sqrt{AS^2 + AB^2} = 4$. In the same fashion, with $\triangle BYC \sim \triangle BDT$,

$$DT = \frac{YC}{BY} \cdot BD = \sqrt{\frac{10}{6}} \cdot 6 = \sqrt{60},$$

hence $CT = \sqrt{DT^2 + CD^2} = 8$. So the sum of the areas of ω_1 and ω_2 is $\pi(2^2 + 4^2) = \mathbf{20\pi}$.

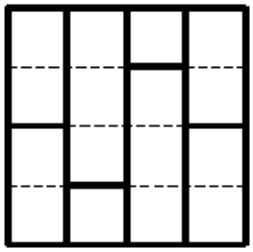
Problem 10. In the number puzzle below, clues are given for the four rows, each of which contains a four-digit number. Cells inside a region bounded by bold lines must all contain the same digit, and each of the eight regions contains a different digit. The variables in the clues are all positive integers. Complete the number puzzle.

1: $4^a + 13^b + 14^c$

2: $5^p + 13^q + 17^r$

3: $4^x + 5^y + 31^z$

4: the average of the other 3 rows



Solution 10. The following solution only relies on the clues for the third and fourth rows. Label the digits in the array as follows.

A	B	C	D
A	B	Y	D
W	B	Y	Z
W	X	Y	Z

The solution proceeds in three steps.

Step 1. The final row is the average of the first three rows, which implies

$$3(1000W + 100X + 10Y + Z) = 1000(2A + W) + 100(3B) + 10(C + 2Y) + (2D + Z).$$

This rearranges to

$$0 = 2000(A - W) + 300(B - X) + 10(C - Y) + 2(D - Z).$$

By comparing the magnitudes of $|2000(A - W)|$ to $|300(B - X) + 10(C - Y) + 2(D - Z)|$, it is impossible to have $|A - W| \geq 2$, hence $A - W = \pm 1$. A similar argument gives $B - X = \mp 7$, and then $C - Y = \pm 9$, $D - Z = \pm 5$, where the signs correspond. In particular, $\{C, Y\} = \{0, 9\}$.

Step 2. Next, the number in the third row satisfies

$$4^x + 5^y + 31^z \equiv 4^x + 6 \pmod{10}$$

and consequently either $Z = 0$ or $Z = 2$. However, one of C and Y must be 0, hence $\boxed{Z = 2}$. This can only occur if $\boxed{D = 7}$, which determines all the signs in the previous step: it follows that $A - W = -1$, $B - X = -7$, $C - Y = 9$, $D - Z = 5$. In particular, $\boxed{C = 9}$ and $\boxed{Y = 0}$.

Step 3. Because $B - X = -7$ and the digits 0 and 2 have already been used, it follows that $\boxed{B = 1}$ and $\boxed{X = 8}$. The third clue now reads

$$4^x + 5^y + 31^z = \underline{W}\ \underline{1}\ \underline{0}\ \underline{2}.$$

The mod 10 calculation in the previous step implies that x is even and in particular, $x > 1$. Taking the previous equation modulo 8 gives $5^y + 31^z \equiv 102 \equiv 6 \pmod{8}$, which can only occur if y is odd and z is even. This implies $z = 2$ (as $z \leq 2$). Therefore $4^x + 5^y \equiv 141 \pmod{1000}$ and because $x \leq 6$ is even, $y \leq 5$, and this forces $x = 2$ and $y \in \{3, 5\}$. The value $y = 3$ would give $W = 1$, which is not permitted because $B = 1$, so $y = 5$ and $\boxed{W = 4}$. Finally, $\boxed{A = 5}$.

The completed number puzzle reads as follows.

5	1	9	7
5	1	0	7
4	1	0	2
4	8	0	2